

# ICBO 2026 - Talks by Day

Author(s) and title for each scheduled talk, grouped by day

## Day 1

### SESSION 1: Bilevel Optimization with Multiple Leaders or Followers (4 Talks)

| Author(s)                                         | Title                                                                                                                                                       |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hori, Atsushi; Okuno, Takayuki; Fukuda, Ellen H.  | Utilization of the Regularized Nikaido-Isoda Function for Multi-Leader-Follower Games Beyond Followers' Convexity                                           |
| Svensson, Anton; Salas, David                     | Existence of solutions for deterministic bilevel games under a general Bayesian approach                                                                    |
| Yu, Xinyue; Liu, Shaonan; Yih, Yuehwen; Kong, Nan | Government Subsidy Optimization for Redesigning Mutually Disjoint for-Profit Service Networks – A Single-Leader Multi-Follower Bilevel Programming Approach |
| Flocco, Dominic; Gabriel, Steve; Schiewe, Philine | A Nested Benders Decomposition Framework for Multi-Follower Bilevel Optimization                                                                            |

### SESSION 2: Robust and Stochastic Problems (4 Talks)

| Author(s)                                                               | Title                                                                                                                                            |
|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Chen, Shuyi; Zhu, Shixiang; Sioshansi, Ramteen                          | Large-Scale Resilience Planning for Wildfire-Prone Electricity-System via Adaptive Robust Optimization                                           |
| Gu, Wei; Chen, Shuyi; Zhu, Shixiang; Sioshansi, Ramteen                 | Multistage Adaptive Robust Optimization for Irreversible Long-term Resilience Planning                                                           |
| Ebrahimi, Mohammadjavad; Shanbhag, Uday V.; Yousefian, Farzad           | On the Resolution of Stochastic MPECs over Networks: Distributed Implicit Zeroth-Order Gradient Tracking Methods                                 |
| Soto-Vergel, Angelo; Walteros, Jose; Ramirez-Rios, Diana; Borrero, Juan | A Bilevel Stochastic Optimization Framework for Strategic Deployment of Survivor Geolocation Systems and Rescue Teams in Post-Disaster Operation |

### SESSION 3: Bilevel Optimization for the Electricity Sector 1 (4 Talks)

| Author(s)                                                                                         | Title                                                                                              |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Chen, Yize                                                                                        | Learning-Augmented Stackelberg Games for Scalable Demand Response in Electricity Markets           |
| Cai, Chengyi; Zhang, Peter; Zhu, Woody                                                            | Two-Stage Stochastic Optimization for Reliable and Grid-Compatible Expansion of AI Data Centers    |
| Bulaich Mehamdi, Abdellah; van Ackooij, Wim; Brotcorne, Luce; Gaubert, Stéphane; Jacquet, Quentin | Electricity Pricing: A Bilevel Programming Approach for Time-of-Use Pricing and Flexible Consumers |
| Rojo, Luis; Anjos, Miguel F.; Brotcorne, Luce; Gendreau, Michel                                   | The Charging Facility Location-and-Operation Problem                                               |

## SESSION 4: Solvers and Instances (3 Talks)

| Author(s)                                                                        | Title                                                                                                                             |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Thürauf, Johannes; Kleinert, Thomas; Ljubić, Ivana; Ralphs, Ted; Schmidt, Martin | BOBILib: Bilevel Optimization (Benchmark) Instance Library                                                                        |
| Bolusani, Suresh; Battista, Federico; Ralphs, Ted                                | GUMBO: A Software Framework for Generalized Benders' Decomposition of Mixed-Integer Bilevel and Two-Stage Stochastic Optimization |
| Ashraf, Mahmoud; Lamperski, Jourdain; Prokopyev, Oleg                            | Generalizing single-level relaxations for bilevel linear programs                                                                 |

## SESSION 5: Nonlinear Bilevel Optimization (3 Talks)

| Author(s)                                                   | Title                                                                                |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Song, Zitao; Padmanabhan, Swati; Zhang, Zhe                 | Goldstein Piecewise Smooth Exploration for Linearly Constrained Bilevel Optimization |
| DEY, SOUMITRA                                               | A New Class of Convex Functions and Characterizations                                |
| Sharifi, Sina; Yazdandoost Hamedani, Erfan; Fazlyab, Mahyar | Sequential QCQP for Bilevel Optimization with Line Search                            |

## SESSION 6: Location Problems (3 Talks)

| Author(s)                                                        | Title                                                                                                          |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Almeida Silva, Warley; Carvalho, Margarida; Jena, Sanjay Dominik | Stackelberg Dynamic Location Planning under Cumulative Demand                                                  |
| Ju, Jiachen; Zeng, Bo; Zeng, Shangzhi; Zhang, Jin                | Solving A Discrete Bilevel Healthcare Facility Location Problem via Submodular Maximization and Branch-and-Cut |
| Choi, Yeonjun; Zeng, Bo; Lee, Kangbok                            | Exact Algorithms for Robust Container Pre-Marshalling and Relocation under Retrieval-Priority Uncertainty      |

## SESSION 7: Uncertainty in Bilevel Optimization 1 (3 Talks)

| Author(s)                                                     | Title                                                                                                          |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Cotrina, John; Flores, Gonzalo; Salas, David; Svensson, Anton | Stability of expectation with decision-dependent uncertainty over moving sets                                  |
| Vasquez, Sebastian; Bodur, Merve; Pagnoncelli, Bernardo       | Bridging Stochastic and Bilevel Optimization: A Unified Framework for Hierarchical Decisions under Uncertainty |
| Xie, Yu; Ralphs, Ted K; Prokopyev, Oleg                       | Robustness and Bounded Rationality in Mixed Integer Bilevel Linear Optimization                                |

## SESSION 8: Bilevel and Inverse Optimization 1 (3 Talks)

| Author(s)                                      | Title                                                                                                       |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Zhou, Wenbin; Orfanoudaki, Agni; Zhu, Shixiang | Conformalized Decision Risk Assessment                                                                      |
| Wu, Guanting; Zhang, Peter                     | Learning from Experts under Decision Coupling: An Application to Robust Disaster Response                   |
| Holani, Anurag; Gupta, Rishabh; Zhang, Qi      | An inverse mixed-integer optimization framework for learning interpretable models of expert decision-making |

## SESSION 9: Equilibria and Bilevel Optimization (3 Talks)

| Author(s)       | Title                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Gautam, Pankaj  | Forward-backward-forward dynamics for bilevel equilibrium problem                                                 |
| Lin, Yen-Cherng | Set Equilibrium Problems with Bilevel Structures                                                                  |
| Zhu, Quanyan    | Bilevel Cooperative Dynamics and Punctuated Equilibrium: From Tactical Consensus to Strategic Coalition Formation |

## Day 2

## SESSION 10: Mixed-Integer Bilevel Optimization (4 Talks)

| Author(s)                                                           | Title                                                                                                                                                               |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Machado Pacheco, Bruno; Carvalho, Margarida; Yu, Kim                | Valid Inequalities for Binary Bilevel Problems with Lower-Level Graph Reformulations                                                                                |
| Feleke, Flagot; Kassa, Semu                                         | Exact Solution Approach for Bi-level Mixed Integer Linear Fractional Programming Problems and its application in Sustainable Supply Chain for Agricultural Products |
| Vasquez, Sebastian; van Hoes, Willem-Jan; Lozano, Leonardo          | Network Flow-based Cuts for Binary Linear Bilevel Optimization                                                                                                      |
| Titus, William; Lunday, Brian; Robbins, Matthew; Hopkinson, Kenneth | The Network Fortification Problem for Evacuation Operations                                                                                                         |

## SESSION 11: Uncertainty in Bilevel Optimization 2 (4 Talks)

| Author(s)                                 | Title                                                                        |
|-------------------------------------------|------------------------------------------------------------------------------|
| Sinha, Ankur                              | Bilevel Optimization under Uncertainty: Applications in Network Interdiction |
| Tian, Yiqi; Zeng, Bo                      | A Computational Study on Distributionally Robust Optimization                |
| Bienstock, Daniel; Sisson, Blake          | Derisking solutions to optimization problems                                 |
| Beck, Yasmine; Ljubić, Ivana; Yan, Wenjin | Robust Combinatorial Optimization with Decision-Dependent Budget Uncertainty |

## SESSION 12: Selected Applications of Bilevel Optimization 1 (3 Talks)

| Author(s)                                    | Title                                                                                                 |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Whitehead, Nathan                            | Network Interdiction of Disaster Response for Rapid Distribution and Maximal Covering of Demands      |
| Zhu, Xiaoyue; Adulyasak, Yossiri; Xia, Wenyi | Joint Pricing and Capacity Investment Problem for Coopetitive Port Authorities                        |
| Sahu, Rishabh                                | When Priority Fails: A Bilevel Framework for Resolving Passenger-Freight Conflicts in Indian Railways |

### SESSION 13: Uncertainty in Bilevel Optimization 3 (4 Talks)

| Author(s)                                                | Title                                                                                                         |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Hao, Hao; Brennan, Mark; Zhang, Peter                    | Protecting Senegal's Road Network Against Flood Risk and Policy Uncertainty                                   |
| Sahu, Vishnu; Zhang, Qi                                  | Robust sequential bilevel programming with unknown follower's objective                                       |
| Perez Tezoco, Jaime; Borrero, Juan S.; Lozano, Leonardo  | A Bilevel Optimization Approach for a Class of Combinatorial Problems with Correlated Disruptions and Probing |
| Liu, Suri; Zhang, Yiling; Basciftci, Beste; Wang, Wenyan | Stochastic Bilevel Optimization for the Network Design of Multimodal Transit Systems                          |

### SESSION 14: Selected Applications of Bilevel Optimization 2 (4 Talks)

| Author(s)                            | Title                                                                                   |
|--------------------------------------|-----------------------------------------------------------------------------------------|
| Deng, Haoyun; Wei, Peng; Xie, Weijun | Bilevel Risk-Averse Routing and Resource Allocation for Urban Air Mobility              |
| Miao, Yidi; Han, Eojin; Zhang, Peter | Joint Spatial Partitioning and Dispatch via Bilevel Optimization for Last-Mile Delivery |
| Sapirstein, Abel                     | MOMS: A Model for Optimizing Maternal Systems of Risk-Appropriate Care                  |
| Song, Mingyao                        | Bi-Level Predict-then-Optimize for Contract Interactions in Livestream Shopping         |

### SESSION 15: Bilevel and Inverse Optimization 2 (4 Talks)

| Author(s)                                   | Title                                                                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ley, Eva; Merkert, Maximilian               | On Integral Inverse Optimization Problems                                                             |
| Nadi, Sina; Lee, Taewoo; Prokopyev, Oleg A. | On the inverse optimal value problem for a class of bilevel linear programs                           |
| Seidel, Tobias                              | Solving inverse robust optimization problems with bilevel-accelerated adaptive discretization methods |
| Ota, Matheus; Vasquez, Sebastian            | A Geometric Perspective on Inverse Combinatorial Optimization via Adjacency                           |

### SESSION 16: Bilevel Optimization and AI Systems (3 Talks)

| Author(s)                                                         | Title                                                                                                                                   |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Dumouchelle, Justin; Julien, Esther; Kurtz, Jannis; Khalil, Elias | Neur2BiLO: Neural Bilevel Optimization                                                                                                  |
| Zhu, Quanyan                                                      | Generative LLM Signaling Games: Censorship Evasion, Semantic Blindness, and Awareness Shaping                                           |
| Garee, Isaac; Giovannelli, Tommaso                                | A stochastic algorithm for bilevel optimization with a mixed-integer upper-level problem and applications to neural architecture search |

### SESSION 17: Learning and Bilevel Optimization 1 (3 Talks)

| Author(s)                                                  | Title                                                                                   |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Zhao, Tongke; Ghadimi, Saeed; Wolkowicz, Henry             | Preconditioned Bilevel Optimization                                                     |
| Elrefaei, Jessica; Hua, Kaixun; Tran, Hoang; Borrero, Juan | Graph4BiLO: Graph-Based Value Function Approximation for Bilevel Optimization           |
| Zhang, Yilang; Li, Bingcong; Giannakis, Georgios B.        | Learnable Loss Geometries with Mirror Descent for Scalable and Convergent Meta-Learning |

### SESSION 18: Bilevel Optimization for the Electricity Sector 2 (3 Talks)

| Author(s)                                               | Title                                                                                                                                                     |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nguyen, Dai Nam; Bhuiyan, Tanveer; Zeng, Bo; Chen, Yang | A Stochastic Game-Theoretic Optimization Approach for Managing Coupled Local Electricity and Thermal Markets with Electric Vehicles and Renewable Sources |
| Andrew, Mastin                                          | A Formulation for Power Flow Interdiction with Bounded Big-M Values                                                                                       |
| Kazi, Saif; Sundar, Kaarthik                            | On the Choice of Single-Level Reformulations for Bilevel Optimization in Optimal Power Flow Networks                                                      |

## Day 3

### SESSION 19: Selected Applications of Bilevel Optimization 3 (4 Talks)

| Author(s)                                                                                   | Title                                                                                                             |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Arango Marin, Jaime; Bustamante Gutierrez, Lina Clemencia; Sierra Narvaez, Francisco Javier | Advanced time series modeling for forecasting imbalances in maritime transport using bi-level optimization        |
| Sheldon, Andrew; Lunday, Brian; Caballero, William                                          | The Location Set Covering Disruption Problem                                                                      |
| Vo, Dung; Tran, Dat; Khanduri, Prashant                                                     | Bilevel Optimization under Non-Convex Non-Smooth Settings                                                         |
| Manimangalam, Praveen                                                                       | A Bilevel Optimization Framework for Self-Configuring Agentic CRM Architectures in Small-Scale Retail Enterprises |

### SESSION 20: Uncertainty in Bilevel Optimization 4 (4 Talks)

| Author(s)                                       | Title                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hao, Hao; Zhang, Peter                          | Robust Paths: Geometry and Computation                                                                |
| Thürauf, Johannes; Lefebvre, Henri              | Robust Bilevel Optimization with a Wait-and-See Follower: A Column-and-Constraint Generation Approach |
| Beck, Yasmine; Büsing, Christina; Ljubic, Ivana | Exact Methods for Solving k-Delete Recoverable Robust 0-1 Problems Under Budgeted Uncertainty         |

## SESSION 21: Interdiction Problems (4 Talks)

| Author(s)                                                | Title                                                           |
|----------------------------------------------------------|-----------------------------------------------------------------|
| Huntington, Rie; Zeng, Bo                                | Bilevel Mixed-Integer Interdiction of the Shortest Path Problem |
| Truong, Tran Mai Anh; Lozano, Leonardo; Sefair, Jorge A. | Network Interdiction with General Integer Variables             |
| Dadpour, Amir; Fukasawa, Ricardo; Wenginger, Noah        | Improved Algorithms for Shortest-Path Network Interdiction      |
| Zhou, Bo; Jiang, Ruiwei; Shen, Siqian                    | Submodularity and Supermodularity in Min-Cost Flow Interdiction |

## SESSION 22: Single-Level Reformulations (3 Talks)

| Author(s)       | Title                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------|
| Zemkoho, Alain  | True single-level reformulation for pessimistic bilevel optimization                                                      |
| Singh, Vinay    | Optimality Conditions for Nondifferentiable Interval-Valued Bilevel Optimization Problems via Tangential Subdifferentials |
| Clemente, Eneko | Dimensionality Reduction in Bilevel Linear Programming                                                                    |

## SESSION 23: First-Order Methods (3 Talks)

| Author(s)                                                     | Title                                                                                               |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Chen, Lesi                                                    | First-Order Bilevel Optimization: Computability, Lower Bounds, and Near-Optimal Algorithms          |
| Bohne, Jason; Rosenberg, David; Kazantsev, Gary; Polak, Pawel | Bregman Geometry for Stochastic Online Bilevel Optimization                                         |
| Ji, Kaiyi                                                     | Lower Complexity Bounds for Nonconvex-Strongly-Convex Bilevel Optimization with First-Order Oracles |

## SESSION 24: Structural and Multiobjective Perspectives in Bilevel Optimization (3 Talks)

| Author(s)                                       | Title                                                                         |
|-------------------------------------------------|-------------------------------------------------------------------------------|
| Chen, Lisha                                     | Optimization over the Pareto set beyond convex objectives                     |
| Ralphs, Ted; Fallah, Samira; Battista, Federico | The Relationship Between Bilevel Optimization and Multiobjective Optimization |
| Weninger, Noah; Fukasawa, Ricardo               | General framework for relaxing some Stackelberg zero-sum games                |